Transformers Find the Item but Misread It: Locating versus Readout in Working Memory

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Abstract

Large language models fail the N-back task in a human-like way: accuracy falls steeply as N grows, and most errors are false alarms on near-target lures. A prominent account attributes this limit to the precision of softmax attention, which must single out one past position among many. We test that account directly in 15 pretrained models (Pythia 70M–6.9B, GPT-2 124M–1.5B, Qwen2.5 0.5B–14B) and in minimal transformers trained from scratch. We separate each error into a *locating* failure (attention lands on the wrong item) and a *readout* failure (attention lands on the right item but the answer is still wrong). Three results favour readout. First, the heads that carry the task attend to the target on about 90% of misses. Second, linear probes on the residual stream separate true matches from lures almost perfectly (median AUC 0.98–1.00) while the model's own answer barely does (0.58–0.63), including on the model's own false alarms. Third, sharpening the identified heads leaves performance unchanged (median $\Delta d' = +0.04$), whereas removing the letter identity of non-target items raises it in 42 of 43 model×N cells (median $+0.60$, up to $+1.78$ in Qwen2.5-14B), five times the gain of a recent content-suppression intervention we replicate. In minimal models, softmax precision does limit a one-layer, residual-free transformer, but depth and a residual stream remove the limit entirely, and even there accuracy tracks how content is used rather than how sharply position is coded. Working-memory limits in transformers are therefore limits of readout under interference, not of attention precision.

1 Introduction

Working memory, the ability to hold and use a few items over seconds, has a famously small capacity in humans [Cowan, 2001, Ma et al., 2014]. The N-back task probes it directly: for each item in a stream, say whether it matches the one N steps back [Kirchner, 1958]. People get worse as N grows, and their errors concentrate on lures, items that match at N±1 [Kane et al., 2007]. Executive-attention accounts explain this as a failure to select the target against competitors [Engle, 2002].

Pretrained language models show the same signature even though the whole sequence sits in their context [Gong et al., 2024, Hu and Lewis, 2025]. Because nothing has been forgotten, the bottleneck must lie in *retrieval*. Two mechanistic accounts compete. The *precision* account holds that softmax attention cannot concentrate enough mass on position $i - N$ when many keys compete, so nearby items leak in [Gong and Zhang, 2024, Veličković et al., 2025]. The *interference* account holds that the model can find the target, but the content of other items is entangled with it in the residual stream and corrupts the decision [Xiong et al., 2026]. The two predict different things: if attention is imprecise, errors should come with attention in the wrong place and sharpening attention should help; if readout is the bottleneck, errors should occur with attention in place, the needed information should be present in the representation, and removing competing content should help.

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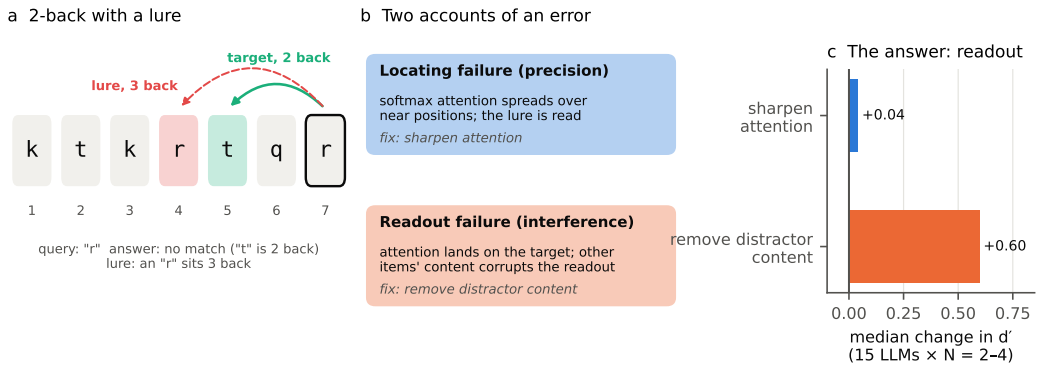


Figure 1: **Overview.** (a) N-back with a lure: the target is the item N back; a lure repeats the query letter at N±1. (b) Two accounts of an error. (c) Across 15 LLMs and N = 2–4, sharpening the heads that carry the task barely changes d', while removing the letter identity of non-target items raises it substantially (medians; Section 7).

We test both predictions with a common set of stimuli, thresholds and interventions across 15 pretrained models and a family of minimal transformers. Our contributions:

- **A locating-versus-readout decomposition** (Section 6). Content-addressed heads place attention on the target line on about 90% of misses; lure false alarms come from heads landing on the same-letter lure. Probes show that whether a match sits at $i - N$ or at a lure offset is linearly decodable in the residual stream, and remains so on the model's own false alarms.
- **A causal contrast on one axis** (Section 7, Fig. 5). Ablating the identified heads costs a median 0.96 d'; sharpening them gains at most 0.04; removing distractor content gains 0.60, with sequence-level bootstrap intervals for every cell.
- **Where precision binds** (Section 5). We show when the softmax precision bound constrains capacity (one layer, no residual stream) and why depth removes it, and find that even there accuracy tracks the use of content, not the sharpness of position.
- **Architecture and development** (Section 8). Fixed-state recurrent models with enough state solve the task, while a selective state-space model declines with N like attention; N-back heads appear early in Pythia training.

2 Related work

Working memory in LLMs. Gong et al. [2024] measured N-back capacity in ChatGPT and found human-like declines; Gong and Zhang [2024] trained small transformers on N-back and related the decline to attention entropy. Hu and Lewis [2025] showed that weaker models drift to an easier task, a confound we control by reporting the best-fitting k for every cell. Zhang et al. [2024] relate working-memory tests to reasoning limits. Closest to us, Xiong et al. [2026] show that LLMs fail working-memory tasks that a two-layer transformer solves, attribute the failure to interference among entangled item representations, and report gains from suppressing letter-identity directions, selected as the best configuration of a sweep on the evaluation trials. We replicate their intervention with held-out selection and compare it directly with head sharpening and with targeted distractor suppression.

Softmax precision. Softmax attention disperses as the number of keys grows [Veličković et al., 2025, Hayase and Karakida, 2026]; scalable softmax [Nakanishi, 2025] and sigmoid attention [Ramapuram et al., 2025] change this behaviour. Expressivity results bound what one attention layer can compute [Hahn, 2020, Sanford et al., 2023]. We use these ideas to state the N-back-specific bound and test whether it is the binding constraint.

Mechanistic interpretability. We build on circuit analyses [Elhage et al., 2021, Wang et al., 2023], induction heads [Olsson et al., 2022], retrieval heads [Wu et al., 2025], and activation and attribution

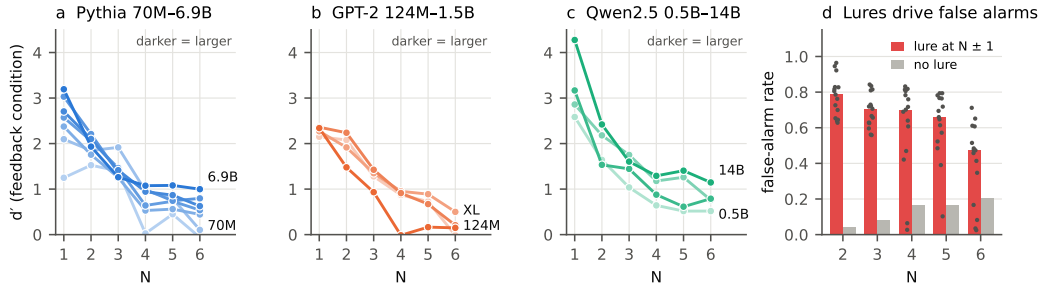


Figure 2: **Behaviour.** (a–c) d' versus N for three model families; darker lines are larger models. (d) False-alarm rates on lures at N$\pm$1 (red; dots are models) and on non-lure items (gray), medians over 15 models.

patching [Zhang and Nanda, 2024, Kramár et al., 2024]. Recurrent and state-space baselines follow Hochreiter and Schmidhuber [1997], Orvieto et al. [2023], Gu and Dao [2024], Katharopoulos et al. [2020]; associative recall in such models is studied by Arora et al. [2024], Jelassi et al. [2024].

3 Setup

Task and stimuli. Sequences of 24 letters from a 20-consonant alphabet, with match rate fixed per N and four planted lures per test sequence (a letter repeating the query at N$\pm$1 or another non-target offset, never creating a match). LLM prompts show three solved demonstration sequences, then the test sequence one item per line ("`k -`", "`t m`"), with correct answers shown for earlier items (*feedback* condition). The score at item i is $\log p(\texttt{m}) - \log p(\texttt{-})$ at the letter token. Thresholds are calibrated on 200 held-out sequences per model, N and condition; all metrics use 1,000 test sequences. A chat format following Gong et al. [2024] is used for Qwen2.5-Instruct models (Appendix B).

Measures. Sensitivity d' with the log-linear correction [Hautus, 1995], hit and false-alarm (FA) rates, FA rates split by lure offset, and the best-fitting k (the k-back rule the model's answers fit best) to detect task drift.

Models. Pythia 70M–6.9B [Biderman et al., 2023], GPT-2 124M–1.5B [Radford et al., 2019] and Qwen2.5 0.5B–14B [Qwen Team, 2024]. Minimal models are attention-only transformers trained from scratch to convergence, varied in depth, residual stream, position encoding and attention normalization (Appendix B).

Heads. For each model and N we rank heads by activation patching from a clean to a corrupted run in which the correct answer flips, scoring the fraction of the mean clean–corrupt logit gap that patching one head recovers [Zhang and Nanda, 2024]. One head typically dominates and is shared across N; at N = 2 the top head recovers 16–73% of the gap.

4 Behaviour: the decline and its lures

All 15 models decline steeply with N (Fig. 2a–c; Table 1). Median d' falls from 2.57 at N = 1 (range 1.25–4.28) to 0.91 at N = 4 and 0.52 at N = 6. Scale helps at N = 1 but little beyond: Qwen2.5-14B reaches 4.28 at N = 1 and 1.15 at N = 6. Errors are dominated by lures (Fig. 2d): at N = 2 the median FA rate is 0.79 on lures at N$\pm$1 against 0.04 on non-lure items. Task drift appears only at large N in small models (best-fitting $k < N$ in 14 of 90 base-model cells, all at N$\geq$4). The chat format reproduces the decline in Qwen2.5-Instruct 0.5B–7B (d' from 1.8–2.0 at N = 1 to 0.4–0.7 at N = 6; Appendix C).

5 When does attention precision bind?

Consider the minimal model of Gong and Zhang [2024]: one attention layer, no residual stream and no MLP, with output $o_i = \sum_j a_{ij} v_j$, a linear decision on o_i, and logits $s_{ij} = c(x_i, x_j) + p(i-j)$ (a content and a position term).

Proposition 1 (the readout is a mass test)**.** *In this model the decision is linear in $\sum_j a_{ij} v_j$, and no linear function of a convex combination of letter embeddings can test whether two letters are equal. The match signal must therefore be carried by how the attention distribution differs between match and non-match trials.*

Proposition 2 (softmax dispersion)**.** *If K keys have logits at least Δ below the target's, the target receives mass at least $1/(1 + Ke^{-\Delta})$; keeping it above $1 - \epsilon$ requires $\Delta \geq \log(K/\epsilon)$. A lure shares the target's content, so only the position term separates them, and the lure-to-target mass ratio is $e^{-\delta_P}$ with $\delta_P = p(N) - p(N \pm 1)$.*

Proposition 3 (depth removes the bound)**.** *With two layers, a residual stream and an absolute or rotary position code, a first-layer head can copy x_{i-N} into position i with margin growing in the weight norm, and the equality test can then be computed in the residual stream, outside the softmax. A monotone recency bias (ALiBi, or no position code) cannot single out offset N in one head.*

Proofs are in Appendix A. Figure 3 tests the propositions. The one-layer model declines with N (d' 2.67 to 2.15 over 50 seeds; panel a). Unnormalized sigmoid attention and top-k attention raise and flatten the curve, while scalable softmax does not (panel b), as Proposition 2 predicts for a normalization-driven limit. With two layers and a residual stream the decline disappears: d' is 6.4–7.2 at every N with learned positions, 5.4–5.8 with sinusoidal and 4.5–5.9 with rotary codes, and 7.7–7.8 for a single model trained on all N with a cue (panel c); ALiBi and no-position models fail at every N (d' 1.3–1.8), as Proposition 3 predicts.

Is dispersion the binding constraint even in the one-layer model? Decomposing the logits of 720 trained checkpoints, the position margin δ_P does not shrink as N grows (4.5 at N = 1, 6.7 at N = 5). Across the 300 checkpoints of the main configuration, δ_P predicts neither lure false alarms (Spearman $\rho = -0.03$) nor d' ($\rho = -0.05$; Fig. 3d). What does predict d' is the size of the difference in target attention between match and non-match trials ($\rho = 0.62$; leave-one-N-out $R^2 = 0.37$; panel e), as Proposition 1 implies. The sign of that difference changes with N: every seed attends more to the target on match trials at N = 1, but only 48% do at N = 4 and 18% at N = 6 (panel f); the rest attend away from a matching letter. Consistent with Proposition 1, every post-hoc edit of the attention pattern (an oracle on $i - N$, or removing off-target mass) lowers d', because off-target mass is part of the code (Appendix C). Precision bounds the minimal model, but the decline is shaped by how content is read out.

6 Locating versus readout in LLMs

Do errors come with attention in the wrong place? For every test item we measure the attention of the top-5 heads from the query letter to the whole line (letter, answer, newline) of item $i - N$ and of its neighbours $i - N \pm 1$, and call the item *located* when the target line receives at least as much attention as either neighbour. On correct matches the heads locate the target in a median 88–91% of items; on correct non-matches only about 25%, because nothing at $i - N$ matches: the heads are content-addressed. On misses the target is located just as often: attention is elsewhere on only 15%, 6% and 8% of misses at N = 2, 3 and 4 (Fig. 4a; medians over 13 models, excluding GPT-2 Large, whose top heads attend to none of the three lines). A logistic probe trained on correct trials then reads the answer from the heads' concatenated outputs: the heads' output already pointed to the wrong answer in 41–57% of misses, and to the right answer, lost in later layers, in 29–43%. Lure false alarms are the mirror image: on 91–96% of them the target line loses to a neighbouring line, typically the lure's, which holds the matching letter.

Is the position of the match represented? Matches and lures both contain the query letter, so only its position separates them. A logistic probe (5-fold, split by sequence) reads match versus lure from the query token's residual stream at nine depths (Fig. 4b,c). The model's own score separates them with median AUC 0.63 (N = 2) and 0.58 (N = 3). The probe reaches a median of 1.00 and 0.98 at the best depth; the input embedding is at chance. On the model's own lure false alarms, the probe still calls a median 95% (N = 2) and 86% (N = 3) of items lures. The largest model shows the same gap: in Qwen2.5-14B the model's own score separates matches from lures with AUC 0.81 and 0.64, the probe with 1.00 and 0.997, and the probe calls 99.8% and 95% of the model's lure false alarms lures. The information peaks in middle layers and declines toward the output (N = 3: 0.92 at the last layer), and the top heads' outputs carry less of it (AUC 0.71–1.00) than the residual stream. Only in the

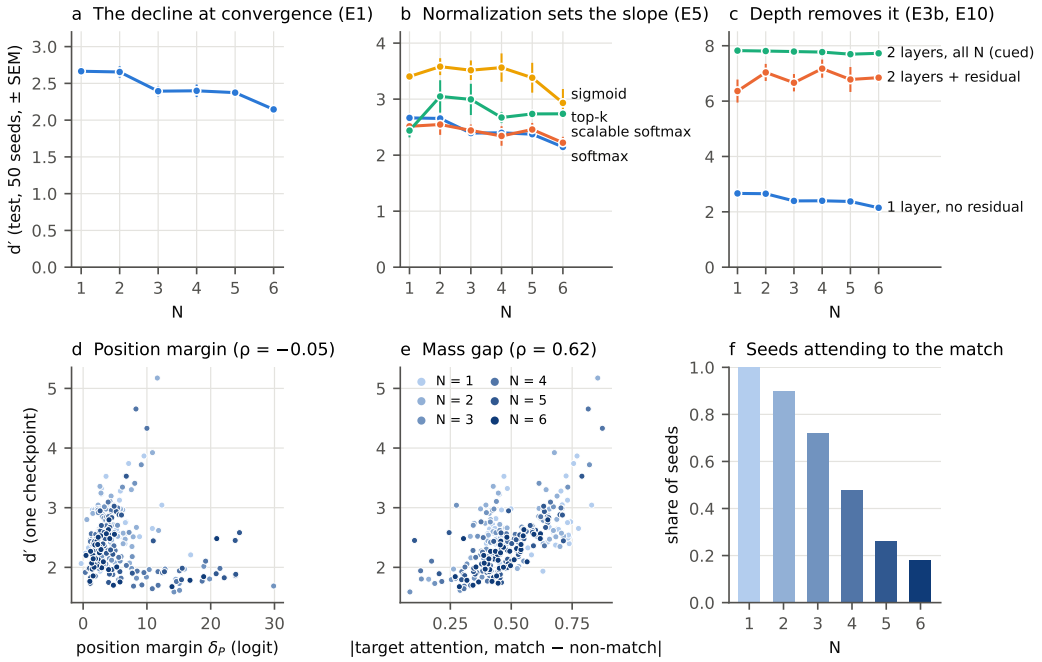


Figure 3: **Minimal models.** (a) The one-layer, residual-free model declines with N at convergence (50 seeds, ±SEM). (b) Sigmoid and top-k attention flatten the decline; scalable softmax does not. (c) Two layers with a residual stream remove it, for one N per model or all N in one cued model. (d, e) Across 300 one-layer checkpoints (50 seeds × N = 1–6; darker = larger N), d' is unrelated to the position margin δ_P but tracks the absolute difference in target attention between match and non-match trials. (f) Share of seeds whose target attention is higher on match trials.

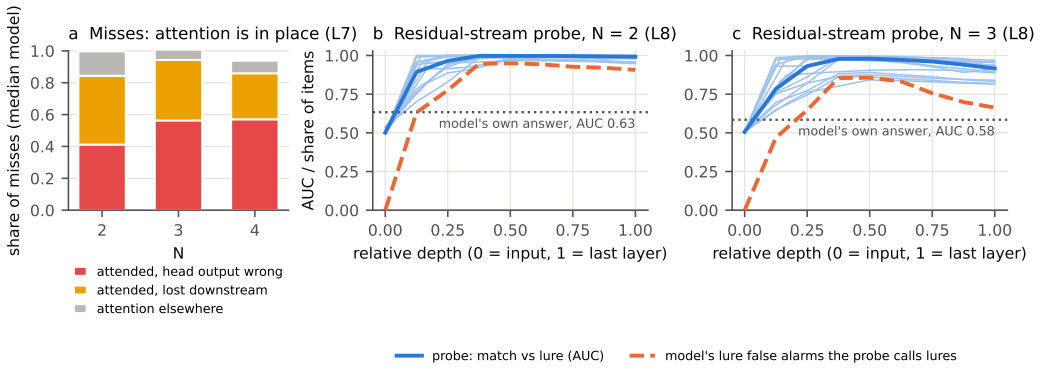


Figure 4: **Locating versus readout.** (a) Misses split by whether the top heads attended to the target line and whether their output carried the right answer (median model). (b, c) A probe separates matches from lures by the position of the matching letter (blue; thin lines are models) and recognizes most of the model's own lure false alarms (orange), far above the model's own answer (dotted).

GPT-2 family at N = 3 is part of the information lost (probe AUC 0.84–0.89). The models compute where the match was, then do not use it.

7 Causal tests: sharpening versus removing interference

We apply five interventions to all 15 models at N = 2–4, on the same sequences, each with its own calibrated threshold (Fig. 5; Table 2). Intervals are 95% paired bootstrap intervals over sequences (2,000 resamples); two degenerate cells whose baseline d' is below 0.3 are excluded.

- **Ablation.** Mean-ablating the top-10 patching heads lowers d' in 39 of 40 cells (median -0.96); ten random heads from the same layers do not (median -0.01). The effect shrinks with scale (Qwen2.5-7B: -0.27 to $+0.03$), indicating more distributed solutions.
- **Sharpening.** Scaling the top-3 heads' logits by $1/\tau$ changes d' by a median $+0.04$ at $\tau = 0.67$ and $+0.03$ at $\tau = 0.8$ (range -0.30 to $+0.30$); flattening them ($\tau = 2$) costs a median 0.38 in the full sweep. Changing the temperature of every head helps most at mild *flattening* ($\tau = 1.25$) in 18 of 26 cells. The heads sit near their optimal sharpness.
- **Distractor suppression.** For each query i, we remove the 19-dimensional letter-identity subspace (fit on calibration sequences at every residual site) from the whole line of every earlier item except $i - N$, scoring each query on its own prefix. At half strength this raises d' in 42 of 43 cells (median $+0.60$, up to $+1.78$), including at N = 4 (median $+0.44$) and in Qwen2.5-14B ($1.60 \rightarrow 3.38$ at N = 3). False alarms fall from about 0.2–0.3 to about 0.1 while hits hold. Removing a random subspace of the same rank does nothing (median -0.03); removing the target's letter destroys performance (median -1.54). Full strength also helps (median $+0.46$) but over-suppresses in Qwen2.5-7B.
- **Replication of Xiong et al. [2026].** One letter-identity direction at one of the two earliest depths, applied at answer positions only, with the configuration chosen on calibration sequences: median $+0.11$ (range $+0.01$ to $+0.30$, 28 cells). The best-of-sweep statistic selected on test data gives $+0.12$, so their effect replicates; it is about a fifth of the effect of targeting the distractor items.

Across cells, distractor suppression and head ablation differ from zero with Wilcoxon $p < 10^{-8}$; sharpening at $\tau = 0.67$ does not ($p = 0.23$). Distractor suppression is oracle-targeted, since it knows which item is the target, so it is a causal test of interference rather than a usable method.

Scale. The two accounts part ways as models grow (Fig. 5b). Ablating the identified heads costs less in larger models (Spearman $\rho = 0.68$ between size and each model's median change, $p = 0.007$, 14 models), from about -1.6 in the smallest Pythia models to -0.27 in Qwen2.5-7B: the solution becomes more distributed. Removing distractor content helps more ($\rho = 0.57$, $p = 0.03$, 15 models). Qwen2.5-14B, the largest model, shows the pattern most clearly. Half-strength suppression raises its d' from 2.42 to 2.88, 1.60 to 3.38 and 1.29 to 2.05 at N = 2, 3 and 4, with every bootstrap interval at least 0.40 above zero; a random subspace of the same rank changes it by -0.10, -0.05 and $+0.18$. With distractor content removed, the model at N = 3 outperforms its own unmodified N = 2 baseline. Scale makes the task less dependent on a few heads, but not less limited by interference.

8 Architectures and development

Recurrent baselines trained on the same stimuli separate two kinds of memory (Fig. 6a,b). LSTM and LRU with a 512-dimensional state solve every N ($d' \geq 7.9$); with a 32-dimensional state the LSTM declines (8.2 to 1.8) while the LRU barely does. Linear attention declines at both sizes, and a selective state-space model, whose gating is input-dependent like attention, declines from 8.2 to 2.9 even with a 512-dimensional state. The decline is not specific to softmax attention; it appears in architectures that retrieve by content-dependent weighting. In Pythia, the top N-back heads and the behaviour appear early in training: d' at N = 2 rises from chance to near its final value between steps 1,000 and 8,000, alongside previous-token and induction heads (Fig. 6c).

9 Discussion and limitations

Pretrained transformers locate the item N back: content-addressed heads land on the target line, and the residual stream encodes whether the matching letter sat at $i - N$ or at a lure position. They fail in readout, where the content of other items interferes with the decision. Sharpening attention cannot fix this; removing the interfering content largely does. The precision account is not wrong

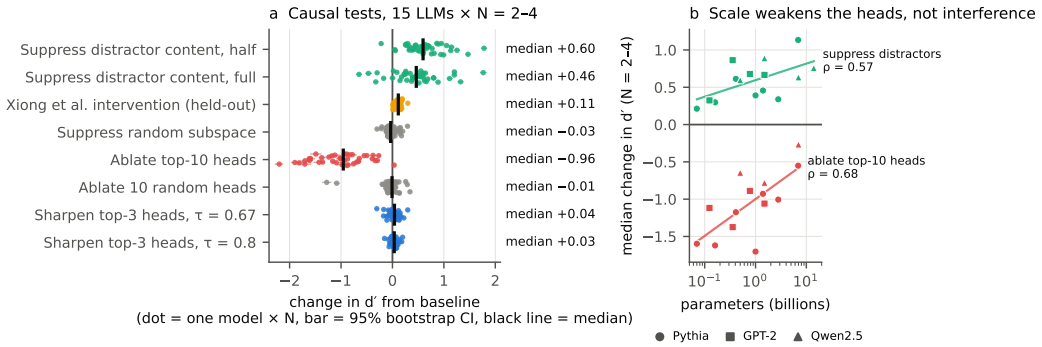


Figure 5: **Causal tests.** (a) Change in d' from baseline for each intervention; each dot is one model $\times$ N cell with its 95% bootstrap interval, black lines are medians. (b) Each model's median change over N = 2–4 against its size, for distractor suppression (half strength) and top-10 head ablation; lines are least-squares fits in log size.

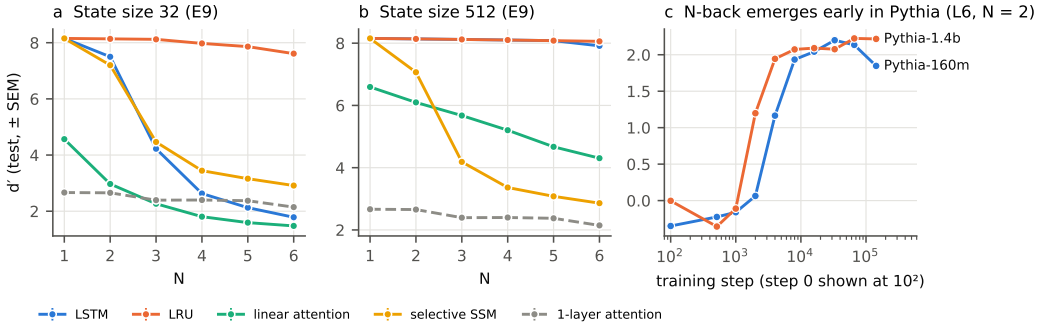


Figure 6: **Architectures and development.** (a, b) Recurrent and linear-attention baselines at state sizes 32 and 512, with the one-layer attention model (dashed) for reference; LSTM-512 overlaps LRU-512. (c) N-back d' over Pythia training (N = 2).

so much as narrow: it describes a one-layer, residual-free model, and depth removes it. This places transformer working memory closer to interference accounts of human working memory than to fixed-slot or attention-bottleneck accounts [Ma et al., 2014, Engle, 2002], and suggests that training or architectural changes that separate item content from the decision variable, rather than sharper attention, are the lever for improving it.

Limitations. We study one task family (letter N-back) with synthetic stimuli. The mechanistic analyses use the line format and the feedback condition. Models range up to 14B parameters, and the identified heads matter less in larger models, so the head-level account may not transfer to frontier models, although the suppression effect grows rather than shrinks with scale in our range (Fig. 5b). Distractor suppression requires knowing the target and is a diagnostic, not a method. The probes show that information is linearly present, not that the model could use it with its existing weights. Our theory is limited to the minimal model.

Broader impact. This is basic research on model mechanisms with public models and synthetic data. Better diagnosis of working-memory failures may help evaluation; the interventions are analysis tools with no deployment use.

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A Proofs

Proposition 1. Let e_x be the value vector of letter x (no residual stream, no MLP, so the output is $o_i = \sum_j a_{ij} e_{x_j}$ up to a fixed linear map, plus positional components that do not depend on the letters). A linear readout computes $w^\top o_i = \sum_j a_{ij} w^\top e_{x_j}$. For fixed attention weights this is additive in the letters, so it cannot represent the indicator $[x_i = x_{i-N}]$, which is not additive (it is an XOR-like function of the two letters). Any dependence of the decision on equality must enter through a_{ij}, which depends on x_i and x_j jointly through the content term $c(x_i, x_j)$. □

Proposition 2. With target logit s_t and K competitors $s_k \le s_t - \Delta$, $a_t = e^{s_t}/(e^{s_t} + \sum_k e^{s_k}) \ge 1/(1 + Ke^{-\Delta})$. Requiring $a_t \ge 1 - \epsilon$ gives $Ke^{-\Delta} \le \epsilon/(1-\epsilon)$, so $\Delta \ge \log(K/\epsilon) + \log(1-\epsilon)$. For a lure ℓ with $x_\ell = x_t$, $c(x_i, x_\ell) = c(x_i, x_t)$ and $a_\ell/a_t = \exp(p(i-\ell) - p(i-t)) = e^{-\delta_P}$. □

Proposition 3 (sketch). With absolute positions P_j and a first-layer head whose query–key product is $\beta\, P_i^\top R P_j$ with R chosen so that $P_i^\top R P_j$ is maximized uniquely at $j = i - N$ with gap $g > 0$, the head places mass at least $1/(1 + Le^{-\beta g})$ on $i - N$; its output, added to the residual stream, makes $e_{x_{i-N}}$ available at position i alongside e_{x_i}, and an MLP or a second attention layer can compute equality of the two letters. Rotary codes provide the same relative-offset selectivity. Under ALiBi, the bias $-m(i-j)$ is monotone in distance, so one head's maximum over positions with equal content is always the most recent, and offset $N > 1$ cannot be singled out. □

B Methods

Minimal models. Attention-only transformers ($d_{\text{model}} = 512$ for the main one-layer configuration, 128 for variants), Adam with learning rate 10^{-4}, batch 32, fresh sequences each step, up to 30,000 steps with early stopping on validation loss (patience 2,000); 50 seeds for the main configuration (E1) and 10 for each variant. Variants: depth and residual stream, learned, sinusoidal, rotary, ALiBi and no position code, softmax, scalable softmax, sigmoid and top-k attention; the cued multi-N model is a two-layer residual model with a cue token giving N. Recurrent baselines (LSTM, LRU, linear attention, selective SSM) use state sizes 32, 128 and 512.

LLM evaluation. Prompts start with a one-sentence instruction and "N = n", then three demonstration sequences with correct answers and the test sequence. Letters, " m" and " -" are single tokens in every tokenizer (checked per model). Models up to 1B run in float32 and larger models in bfloat16. The chat format sends one user turn per letter and one assistant turn per answer.

Head ranking (L3). Clean and corrupted prompts differ only in the letter at $i - N$, flipping the answer. For each head we patch its output from the clean run into the corrupted run at all positions and report the recovered fraction of the mean logit-difference gap (set to missing when the gap is below 0.1).

Locate analysis (L7). Attention from the query letter token to all tokens of a line is summed per head and averaged over the top-5 heads. The answer probe is a logistic regression on the top-5 heads' outputs at the query, trained on up to 4,000 correct items and applied to errors.

Probe (L8). Items with a same-letter item at $i - N$ (matches) or at a lure offset (lures); logistic regression with L_2 penalty on standardized features, five folds split by sequence.

Interventions (L5, B1). Sharpening multiplies a head's attention logits by $1/\tau$; ablation replaces a head's output by its mean over calibration items; random heads are drawn from the same layers. Suppression replaces h by $h - \alpha\,((h - \mu)U)U^\top$ at every residual site (input to the first layer and every layer's output) over the tokens of each non-target line, where U spans the 19 centred class means of the 20 letters at test-item letter tokens. The random control uses a random orthonormal U of the same rank.

Compute. Minimal models mostly on a CPU cluster (about 1,060 core-hours), with some sweeps on single H100 GPUs; LLM studies on single L40S, H100 or B200 GPUs per model and N. The whole project, including preliminary and failed runs, used about 51 GPU-hours.

Licenses. Pythia (Apache-2.0), GPT-2 (MIT), Qwen2.5 (Apache-2.0; Qwen license for some sizes).

C Additional results

Table 1: d' by model and N (line format, feedback condition), and false-alarm rates on N$\pm$1 lures versus non-lure items at N = 2.

Model	N=1	2	3	4	5	6	lure FA, N=2
Pythia-70m	1.25	1.52	1.40	0.03	0.45	-0.06	0.83 / 0.12
Pythia-160m	2.10	1.85	1.92	0.96	0.79	0.10	0.79 / 0.02
Pythia-410m	2.37	1.76	1.25	0.53	0.56	0.44	0.76 / 0.04
Pythia-1b	2.57	2.06	1.46	0.64	0.73	0.54	0.83 / 0.04
Pythia-1.4b	3.03	2.21	1.36	0.99	0.74	0.80	0.73 / 0.02
Pythia-2.8b	2.71	2.10	1.42	0.94	0.87	0.63	0.82 / 0.03
Pythia-6.9b	3.19	1.93	1.26	1.07	1.08	1.00	0.64 / 0.03
GPT-2	2.15	2.08	1.28	0.87	0.73	-0.02	0.96 / 0.08
GPT-2 medium	2.27	1.92	1.35	0.95	0.89	0.50	0.92 / 0.06
GPT-2 large	2.36	2.24	1.43	0.91	0.67	0.21	0.95 / 0.04
GPT-2 xl	2.34	1.48	0.93	-0.01	0.16	0.15	0.70 / 0.06
Qwen2.5-0.5B	2.58	1.64	1.04	0.64	0.52	0.52	0.81 / 0.06
Qwen2.5-1.5B	2.86	2.18	1.75	1.18	1.26	0.76	0.65 / 0.01
Qwen2.5-7B	3.17	1.53	1.44	0.88	0.61	0.79	0.64 / 0.06
Qwen2.5-14B	4.28	2.42	1.60	1.29	1.41	1.15	0.63 / 0.01

Table 2: Change in d' under each intervention; bold entries have 95% bootstrap intervals excluding zero. † degenerate baseline (excluded from summaries).

Model	N	base d'	sharpen	ablate top	ablate rand.	suppress	supp. rand.
Pythia-70m	2	1.52	**-0.11**	**-1.71**	**-1.08**	**+0.38**	**-0.30**
	3	1.40	**+0.22**	**-1.49**	**-1.28**	**+0.05**	**-0.21**
	4	0.03†	**+0.97**	**-0.17**	+0.04	**+0.82**	**+0.31**
Pythia-160m	2	1.85	**+0.14**	**-1.62**	+0.01	**+1.25**	**+0.29**
	3	1.92	**-0.17**	**-1.90**	**-0.13**	**+0.30**	+0.03
	4	0.96	**+0.30**	**-1.09**	**-0.20**	**-0.22**	+0.01
Pythia-410m	2	1.76	**+0.17**	**-1.36**	**-0.03**	**+0.61**	-0.01
	3	1.25	**-0.08**	**-1.18**	**+0.13**	**+0.60**	**-0.04**
	4	0.53	**+0.13**	**-0.39**	**+0.08**	**+0.70**	**+0.05**
Pythia-1b	2	2.06	**-0.04**	**-1.81**	**+0.13**	**+0.59**	**-0.05**
	3	1.46	**+0.11**	**-1.70**	**+0.34**	**+0.26**	**-0.12**
	4	0.64	**+0.09**	**-0.83**	**+0.29**	**+0.39**	**-0.09**
Pythia-1.4b	2	2.21	**+0.08**	**-1.61**	**-0.06**	**+0.95**	**-0.05**
	3	1.36	**+0.05**	**-0.93**	**+0.05**	**+0.46**	**+0.05**
	4	0.99	**+0.05**	**-0.66**	+0.02	**+0.44**	**-0.07**
Pythia-2.8b	2	2.10	**-0.05**	**-1.59**	**-0.12**	**+0.25**	**-0.03**
	3	1.42	**+0.12**	**-1.01**	**-0.03**	**+0.34**	-0.01
	4	0.94	**+0.12**	**-0.70**	**+0.02**	**+0.41**	**+0.05**
Pythia-6.9b	2	1.93	**+0.03**	**-1.02**	**-0.08**	**+0.94**	**-0.08**
	3	1.26	**+0.10**	**-0.34**	**+0.26**	**+1.47**	**+0.13**
	4	1.07	**+0.07**	**-0.55**	+0.03	**+1.13**	**+0.09**
GPT-2	2	2.08	**-0.30**	**-2.20**	**-0.09**	**+0.05**	**-0.22**
	3	1.28	-0.00	**-1.12**	**-0.09**	**+0.71**	**-0.07**
	4	0.87	**-0.02**	**-0.86**	**+0.07**	**+0.32**	**-0.12**
GPT-2 medium	2	1.92	+0.02	**-1.38**	**+0.14**	**+1.05**	+0.00
	3	1.35	**-0.06**	**-1.43**	**-0.10**	**+0.86**	**-0.08**
	4	0.95	**-0.07**	**-0.98**	**-0.18**	**+0.34**	**-0.12**
GPT-2 large	2	2.24	**-0.17**	**-1.16**	**-0.09**	**+0.68**	**+0.06**
	3	1.43	**+0.15**	**-0.89**	**+0.02**	**+0.77**	**-0.03**
	4	0.91	**+0.06**	**-0.50**	**-0.05**	**+0.55**	**-0.03**
GPT-2 xl	2	1.46	**+0.12**	**-1.24**	**-0.06**	**+0.52**	**+0.15**
	3	0.94	**+0.12**	**-0.88**	+0.01	**+0.81**	**-0.05**
	4	-0.02†	**+0.55**	**-0.16**	**+0.53**	**+0.91**	**+0.48**
Qwen2.5-0.5B	2	1.64	**-0.03**	**-0.87**	**+0.06**	**+0.49**	**+0.15**
	3	1.04	**+0.05**	**-0.65**	**-0.11**	**+0.67**	**-0.07**
	4	0.64	-0.01	**-0.38**	+0.02	**+0.60**	**-0.09**
Qwen2.5-1.5B	2	2.18	**+0.10**	**-0.76**	**+0.06**	**+0.89**	**-0.11**
	3	1.75	+0.01	**-0.93**	**-0.05**	**+1.15**	+0.01
	4	1.18	**-0.04**	**-0.78**	**-0.11**	**+0.28**	**-0.09**
Qwen2.5-7B	2	1.53	+0.01	**-0.27**	**+0.06**	**+0.41**	**+0.04**
	3	1.44	**-0.11**	**-0.29**	-0.02	**+0.99**	**+0.20**
	4	0.88	+0.00	+0.03	+0.00	**+0.63**	**+0.14**
Qwen2.5-14B	2	2.42	–	–	–	**+0.46**	**-0.10**
	3	1.60	–	–	–	**+1.78**	**-0.05**
	4	1.29	–	–	–	**+0.76**	**+0.18**

Post-hoc attention edits in the minimal model (E7). On 720 one-layer checkpoints, a fixed split of attention between i and $i - N$ gives $d' \approx 0$ at every N; keeping only the model's own mass on i and $i - N$ lowers d' from 2.70 to 0.63 at N = 1 and from 2.18 to 1.32 at N = 6, with false alarms rising from 0.03 to 0.67 at N = 1. Removing either the content or the position term of the logits gives $d' \approx 0$.

Table 3: Locate (L7) and probe (L8) results per model.

		L7: share of misses			L8: match vs. lure		
Model	N	elsewhere	lost later	head wrong	model AUC	probe AUC	FA called lure
Pythia-70m	2	0.08	0.49	0.43	0.58	0.97	0.75
	3	0.01	0.42	0.57	0.57	0.91	0.61
Pythia-160m	2	0.26	0.34	0.40	0.63	0.98	0.82
	3	0.03	0.59	0.38	0.58	0.99	0.85
Pythia-410m	2	0.11	0.36	0.53	0.66	1.00	0.95
	3	0.06	0.40	0.54	0.58	0.98	0.84
Pythia-1b	2	0.11	0.32	0.57	0.68	1.00	0.95
	3	0.06	0.30	0.64	0.60	0.98	0.87
Pythia-1.4b	2	0.12	0.72	0.16	0.76	1.00	0.99
	3	0.06	0.38	0.56	0.62	0.99	0.92
Pythia-2.8b	2	0.04	0.36	0.60	0.71	1.00	1.00
	3	0.04	0.38	0.58	0.60	1.00	0.96
Pythia-6.9b	2	0.19	0.55	0.26	0.74	1.00	1.00
	3	0.08	0.38	0.54	0.60	1.00	0.98
GPT-2	2	0.16	0.65	0.19	0.49	0.97	0.84
	3	0.12	0.38	0.50	0.48	0.84	0.53
GPT-2 medium	2	0.18	0.53	0.29	0.48	0.99	0.87
	3	0.07	0.39	0.54	0.47	0.86	0.57
GPT-2 large	2	0.72	0.15	0.13	0.56	0.99	0.92
	3	0.54	0.20	0.26	0.50	0.88	0.59
GPT-2 xl	2	0.15	0.42	0.43	0.60	1.00	0.92
	3	0.10	0.28	0.62	0.49	0.89	0.69
Qwen2.5-0.5B	2	0.16	0.50	0.34	0.62	1.00	1.00
	3	0.09	0.23	0.67	0.53	0.99	0.91
Qwen2.5-1.5B	2	0.17	0.42	0.41	0.78	1.00	1.00
	3	0.04	0.39	0.57	0.67	0.99	0.86
Qwen2.5-7B	2	0.14	0.43	0.43	0.63	1.00	0.98
	3	0.45	0.19	0.36	0.60	0.98	0.88
Qwen2.5-14B	2	–	–	–	0.81	1.00	1.00
	3	–	–	–	0.64	1.00	0.95

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